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A Control and Testing Framework for a Digital Micromirror Device in Neutral Atom Quantum Computing and Simulation Platforms

M. Mathews*, X. Yin*, A. Baumgärtner†, T. Esslinger†, A. Akin†
*Department of Information Technology and Electrical Engineering, ETH Zürich, 8092 Zürich, Switzerland
†Department of Physics, ETH Zürich, 8093 Zürich, Switzerland

***Abstract*—Digital micromirror devices (DMDs) have seen recent popularity as spatial light modulators (SLMs) for arbitrary and dynamic potential generation in the manipulation of neutral atom systems for quantum simulation and quantum computing platforms. In enabling the use of a DMD for this purpose, its latency, frame rate, and micromirror movement dynamics must be well characterized for various settings. In this paper, we present our FPGA-based system to control the DMD and the optical setup developed for its verification. Image information and control parameters are generated and configured using a Python user interface and sent to the FPGA using a C++ wrapper and the Cypress FX2 USB protocol. The FPGA design enables the leveraging of the high frame rate capabilities of the DMD as well as ensuring deterministic operation by first storing this USB data on an external DDR memory and updating the micromirror states upon receiving an external trigger pulse. The DMD is imaged onto a scientific camera and two photodiodes as a means of verifying the system behavior and investigating the metrics of interest. Using this framework, the DLP7000 consisting of 768x1024 micromirrors has been shown to refresh at 22 kHz with 45.4 μs latency. Furthermore, a single block consisting of 48x1024 micromirrors is able to refresh at 140 kHz with 7.1 μs latency. The loading and clearing behavior of micromirrors has also been characterized for varying refresh rates.**

***Index Terms*—Digital Micromirror Device, Quantum Simulation, Neutral Atom, FPGA, Quantum Optics**

I. Introduction

The control and manipulation of neutral atoms are a powerful tool to simulate quantum many-body-systems [1], and have a potential for quantum computing [2]. This is typically implemented through the generation and modulation of optical lattices with spatial light modulators (SLMs) to prepare states, encode Hamiltonians, and resolve dynamical properties [3]. Digital micromirror devices (DMDs) have seen recent popularity as SLMs that operate at higher frequencies than relevant alternatives such as nematic liquid crystal displays (LCDs). While DMDs have seen use in some quantum optic experiments, little literature exists on their high-frequency operation (>20kHz) and micromirror movement dynamics [4].

In this work, we present the development of a control framework for an FPGA-based DMD as well as a testing setup used for the verification and validation of this design. The presented FPGA-based system and the DMD timing characteristics results would help the efficient and careful design of quantum optic experiment setups using DMDs.

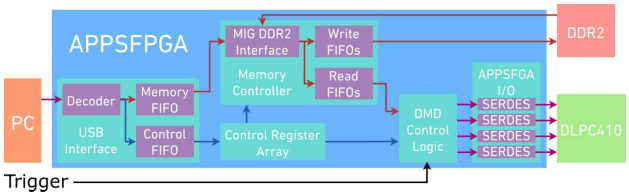

Fig. 1. Proposed FPGA design. Red arrows indicate image data and blue arrows indicate control parameters.

II. FPGA Design

The Applications FPGA (APPSFPGA, Xilinx Virtex 5 LX50) on the DLPLCRC410EVM controller board and the the DMD DLPLCR70EVM are used for our developments. The top-level block diagram of the proposed FPGA architecture is presented in Fig. 1. The basic features for the *USB Interface* and *DMD Control Logic* were already provided by the reference design of Texas Instruments (TI). However, the reference design could reach a maximum 1 Hz refresh rate when the dynamic update of the DMD image is required. We added a memory controller to the reference design and improved the *USB Interface* and *DMD Control Logic* modules to interface with the *Memory Controller*. A maximum of 2543 images can be stored in external 2GB DDR2 SO-DIMM MT16HTF25664HY667E1. The proposed design is only limited by the settling time of the DMD. In addition, we implemented a precise time control feature in the FPGA design using an external trigger to measure the dynamic characteristics of the mirrors.

III. Testing Framework

A complex setup is required for the verification and evaluation of DMD at very high speeds, as presented in Fig. 2. Component F1 represents a collimated laser output producing coherent 532 nm light. Components M1-6 are mirrors used to redirect the path of light. The lenses, L1 and L2, both have a focal length of 300 mm. The components C1 and C2 represent imaging options, each either a scientific camera (BFLY-U3-23S6M) or photodiode (PDA36A2). The beamsplitter PBS1 allows for the laser to reach both the DMD and also collect the reflected DMD image. The second beamsplitter, PBS2, allows us to observe the DMD using multiple imaging tools. First, the functionality of update modes (Global, Quad, Double, or

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TABLE I
A SUMMARY OF THE DIGITAL CONTRIBUTIONS TO THE LATENCY METRIC AS WELL AS THE EXPERIMENTALLY ACHIEVED VALUES

Update Mode	Rows	Memory Read Clock Cycles	Data Load Clock Cycles	Reset Clock Cycles	Total Clock Cycles	Digital Latency (μs)	Experimental Latency (μs)	Refresh Rate (kHz)
Global	768	50	8192	908	9150	45.75	45.35	22
Quad	192	13	2048	908	2969	14.85	14.24	70
Dual	96	7	1024	908	1939	9.70	9.58	104
Single	48	4	512	908	1424	7.12	7.10	140

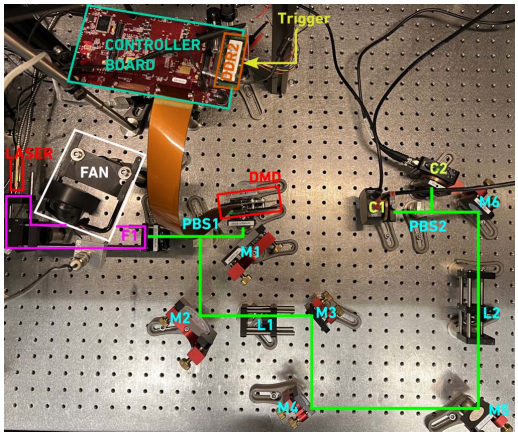

Fig. 2. Optical experiment schematic used to verify and analyze the timing capabilities of the system.

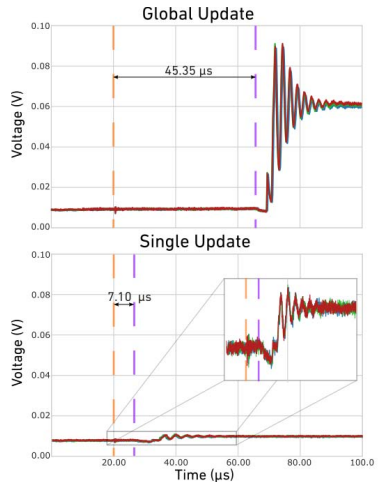

Fig. 3. Photodiode output (red - 100 Hz, blue - 1 kHz, green - 10 kHz) illustrating latency for the global and single update modes with varying external trigger frequencies. The orange line represents the trigger pulse while the purple line indicates the beginning of state change.

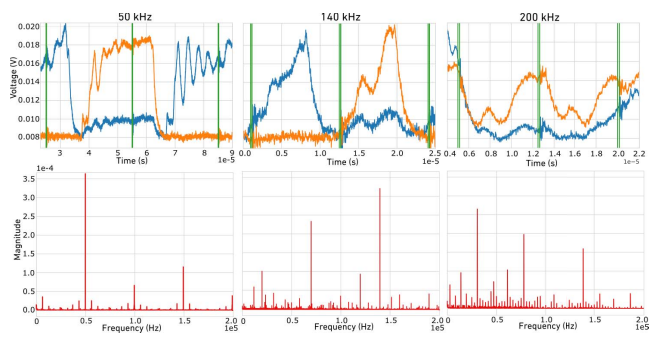

Fig. 4. Photodiode outputs (orange, blue) illustrating high frequency operation and mirror dynamics. Green curves indicate trigger pulses. Red plots indicate fourier transform analysis of the photodiode outputs.

Single) were verified using a camera. The detailed analysis was then performed with photodiodes and an oscilloscope since their acquisition rates are much higher than scientific cameras.

IV. Results and Discussion

Two images are selected to be displayed by the DMD such that they illuminate a single photodiode at a time. The images are uploaded onto the controller board and alternated every time an external trigger is applied.

1) Latency: Latency is defined as the time between a trigger being applied to the system and the observation of the micromirrors changing state, as presented in Fig 3. Table 1 quantifies the digital contributions to the latency metric and compares the experimentally achieved values. Digital latency is measured using ChipScope Integrated Logic Analyzer (ILA) with a clock at 200 MHz. The experimental latency is measured using a Picoscope connected to photodiodes with an external trigger connected to the controller board.

2) Refresh Rate: As the external trigger frequency is increased from 5 kHz to 140 kHz, we observe a decreasing period of time the mirror remains in its steady-state position. At frequencies greater than or equal to 150 kHz however, we no longer observe the asynchronous behavior between the alternating images displayed, as presented in Fig. 4. We observed the maximum refresh rate as 22 kHz in global mode and 140 kHz in single mode.

3) Mirror Dynamics: The investigation of micromirror dynamics concerned the metastable behavior of the micromirrors as they transitioned between states. The plots in Fig. 4 are from the photodiode at the C1 position and with the DMD operating in the single block mode so as to have the largest operating regime in trigger frequency. At frequencies below 35 kHz, we observe the clear turn-on and turn-off behaviour characterized by a series of damped oscillations and a single critically damped oscillation, respectively. For higher frequencies these behaviours begin to merge until entirely dissociating into linear turn-on and exponential turn-off curves at 140 kHz.

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